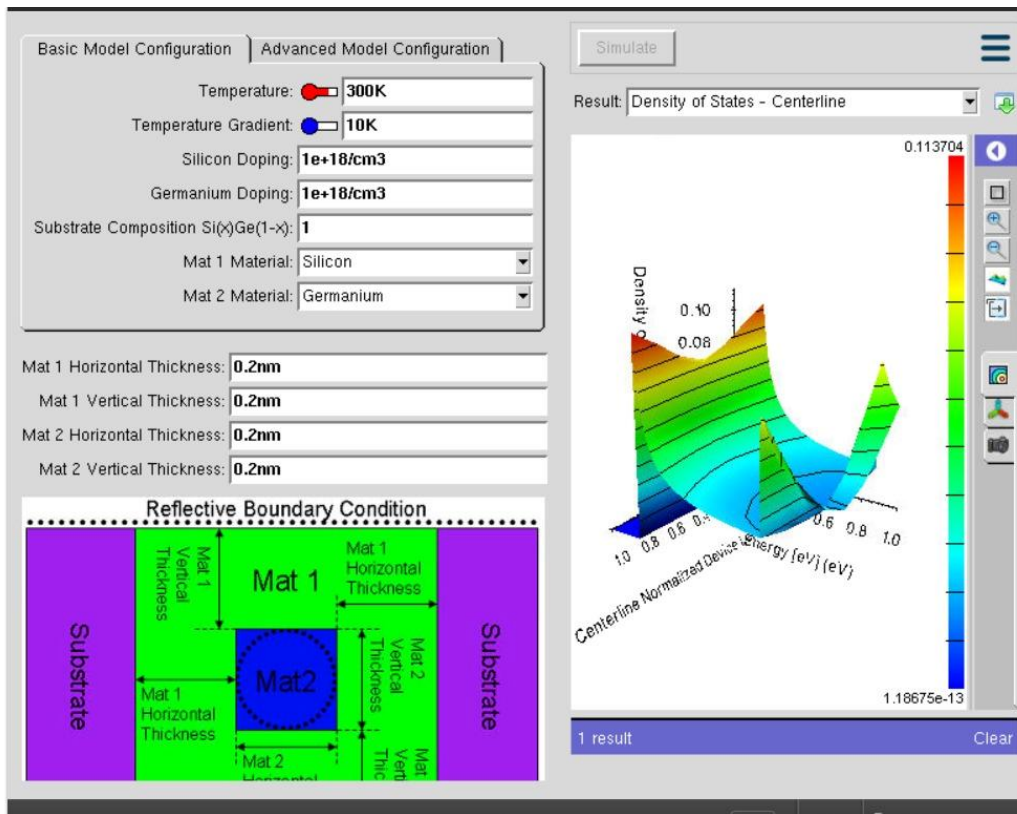
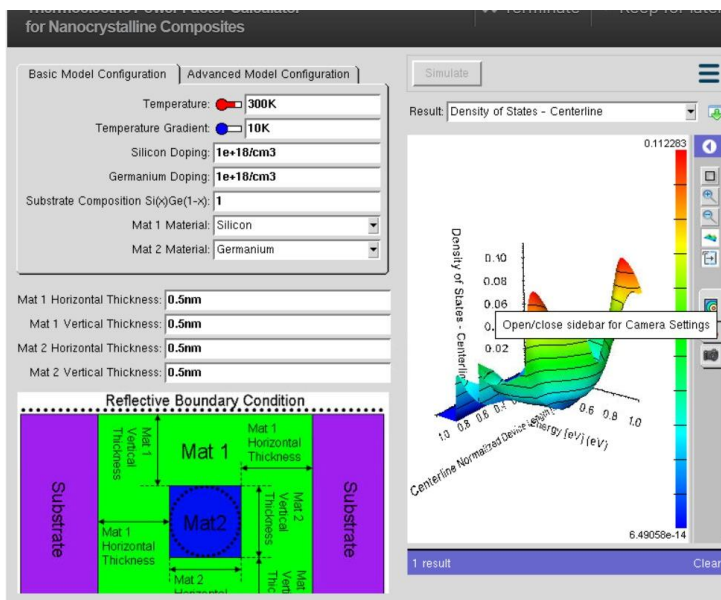


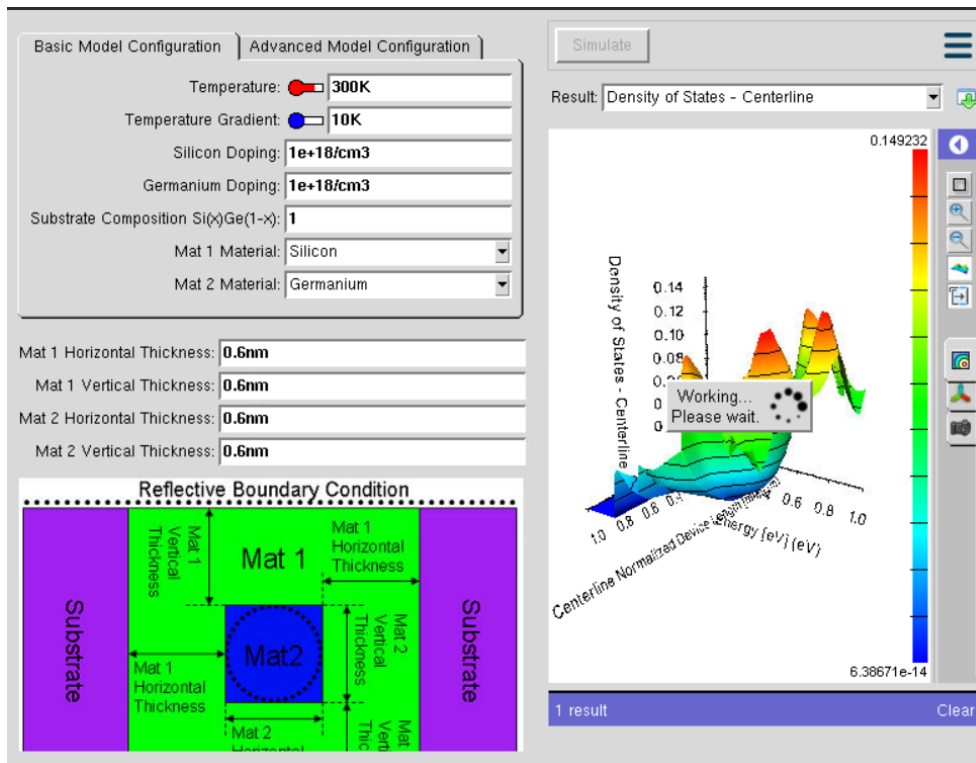
Exp. 2. Simulation of Density of States of Quantum Well Structure with Various Nanometer Thickness.



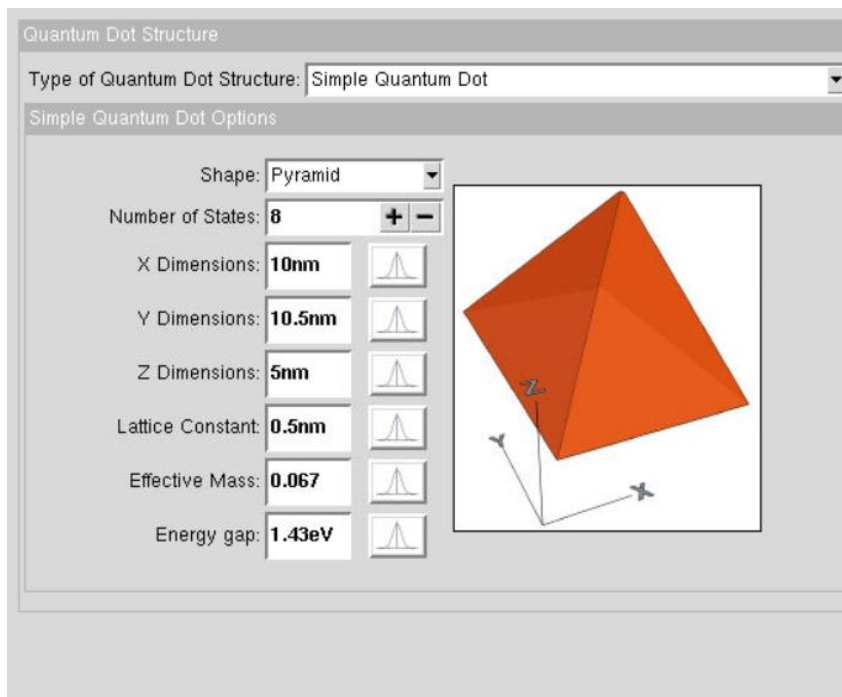
Case Study 1: Simulate the density of states of quantum well of thickness 0.5 nm



Case Study 2: Simulate the density of states of quantum well of thickness 0.6 nm



Exp. 3. Simulation of Energy States of Quantum Dot using NanoHub



Quantum Dot Lab

✕ Terminate

➡ Keep for later

1 Introduction → 2 Structure → 3 Optical → 4 Advanced → 5 Simulate

Light Incident Angles

Angle Theta: 45deg

Angle Phi: 0deg

Sweep

Sweep Parameter: Angle phi (deg)

Minimum: 0

Maximum: 90

Number of Points: 3

Light Source

Absorption

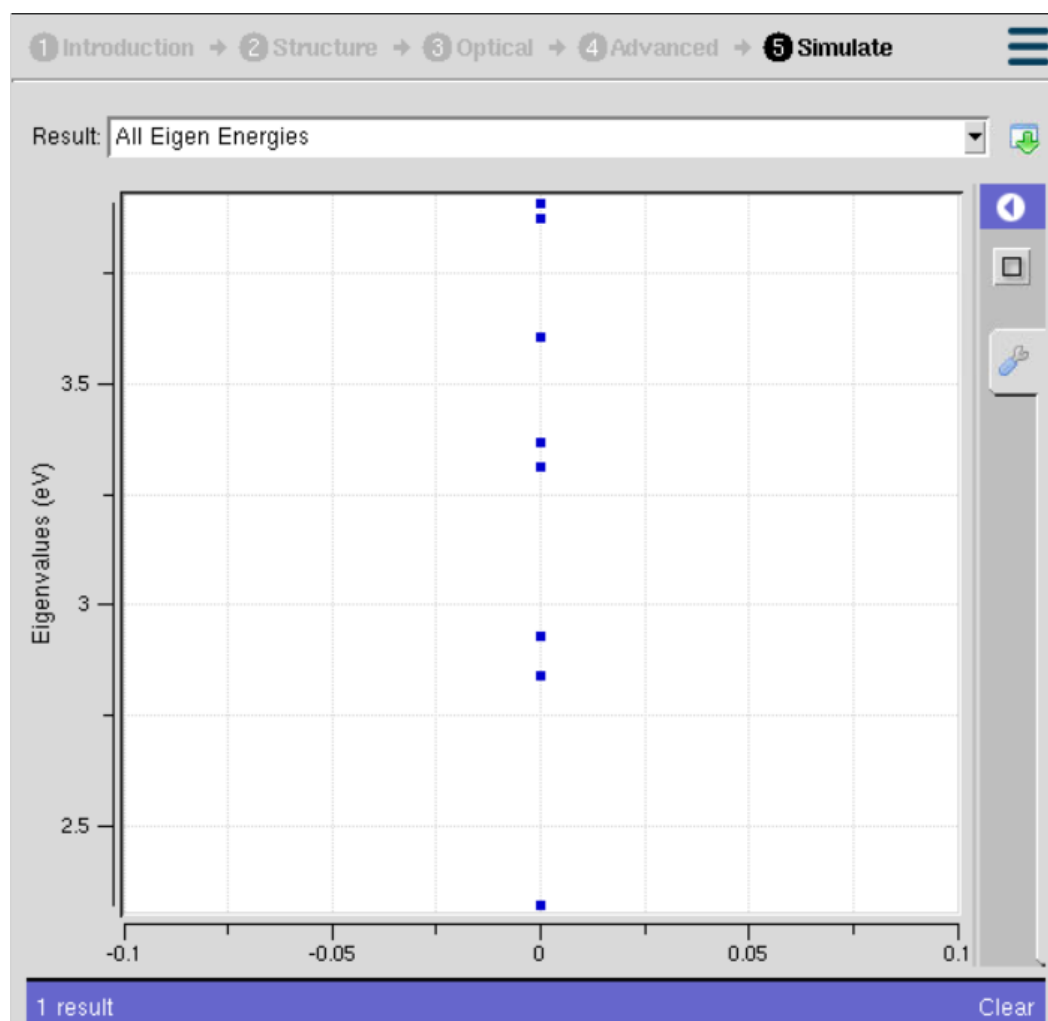
Simple Q Dot - Absolute Fermi Level: ☐ no

Simple Q Dot - Rel. or Abs. Fermi Level: 0eV

Multi-Layer - Fermi Level: 0.7eV

Temperature: ☐ 300K

State Broadening: 0.001



Case Study 1: Simulate the density of states of quantum well of thickness 0.5 nm

Quantum Dot Lab ✕ Terminate ➡ Keep for later

1 Introduction → 2 Structure → 3 Optical → 4 Advanced → 5 Simulate

Quantum Dot Structure

Type of Quantum Dot Structure: Simple Quantum Dot

Simple Quantum Dot Options

Shape: Cylinder

Number of States: 8 + -

X Dimensions: 15nm 

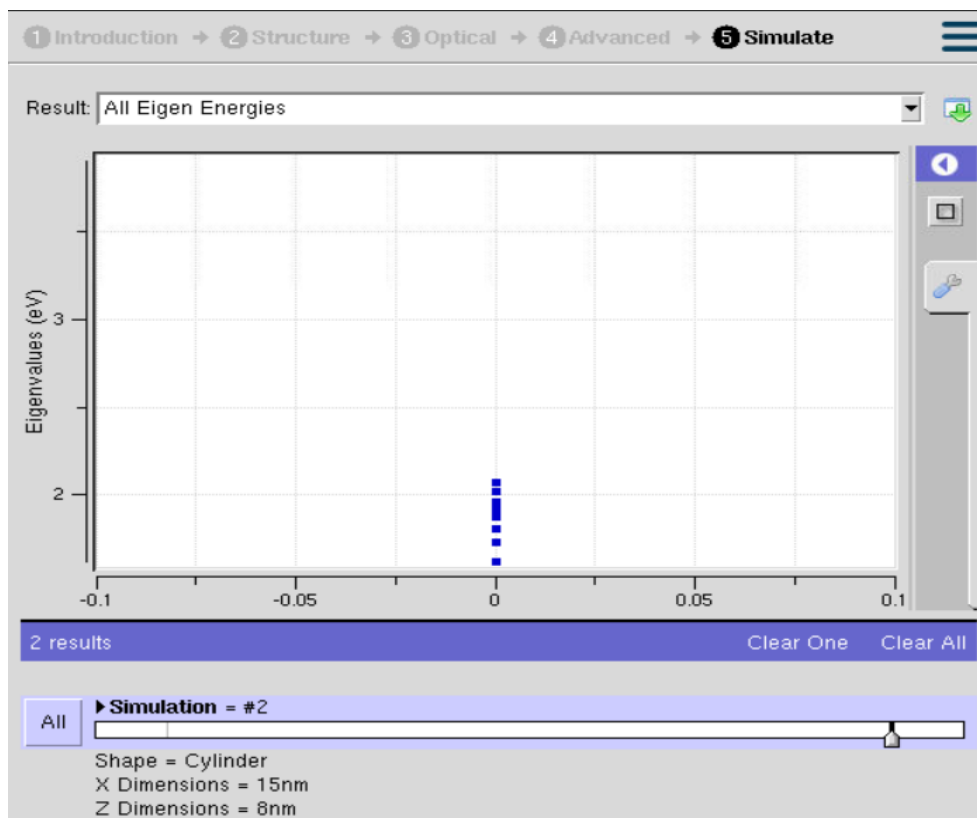
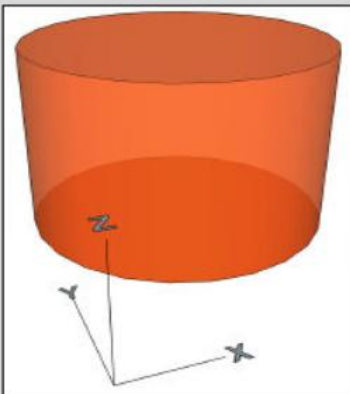
Y Dimensions: 10.5nm 

Z Dimensions: 8nm 

Lattice Constant: 0.5nm 

Effective Mass: 0.067 

Energy gap: 1.43eV 



Case Study 2: Simulate the density of states of quantum well of thickness 0.6 nm

1 Introduction → 2 Structure → 3 Optical → 4 Advanced → 5 Simulate

Quantum Dot Structure

Type of Quantum Dot Structure: Multi-Layer Quantum Dot

Multi-Layer Quantum Dot Options

Shape: Cuboid

Cond. Band States: 6

Val. Band States: 8

Basis: TB10 band 5 orb sp3s_so

Materials: GaAs - InAs - GaAs

Materials: GaAs - InAs - GaAs

Strain: ☐ yes

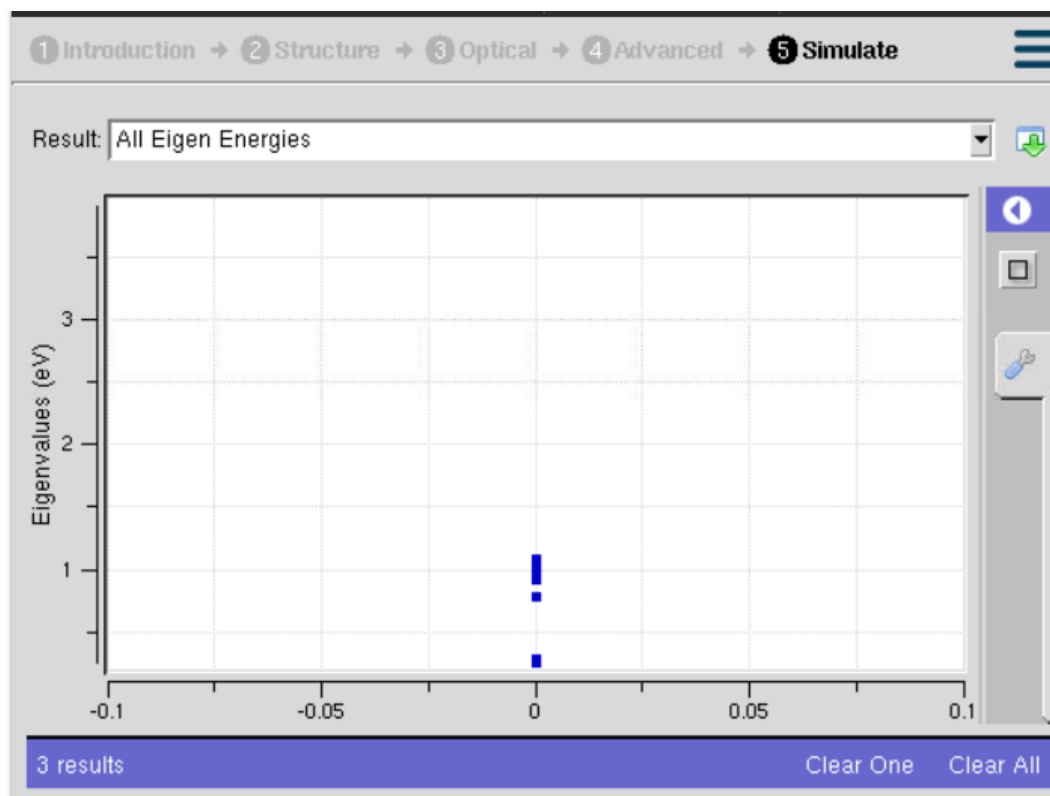
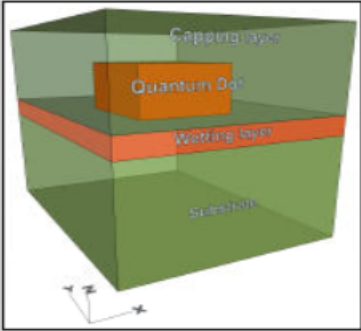
Local Bands: ☐ yes

Quantum Dot Alloy Lateral Domain Vertical Layers

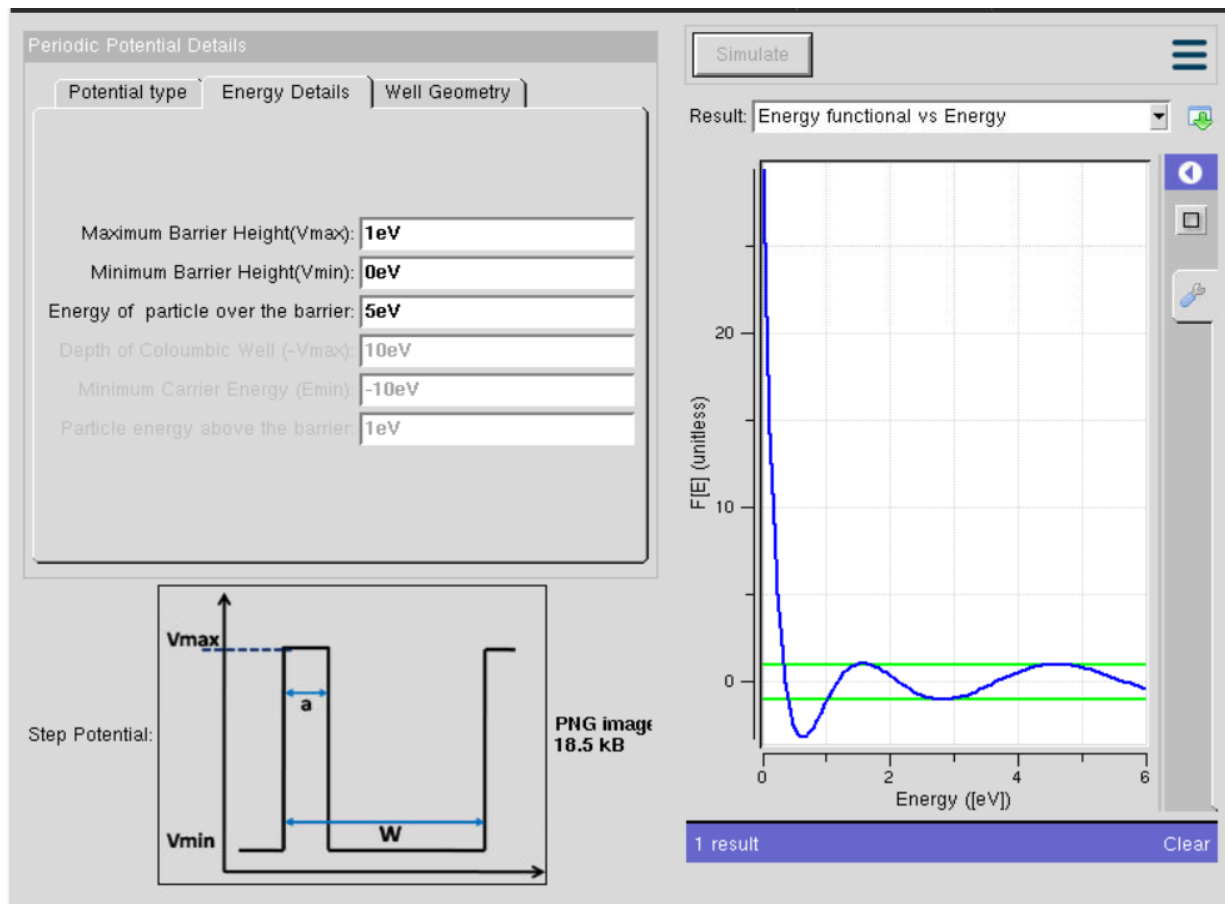
X Dimension: 8nm

Y Dimension: 8.5nm

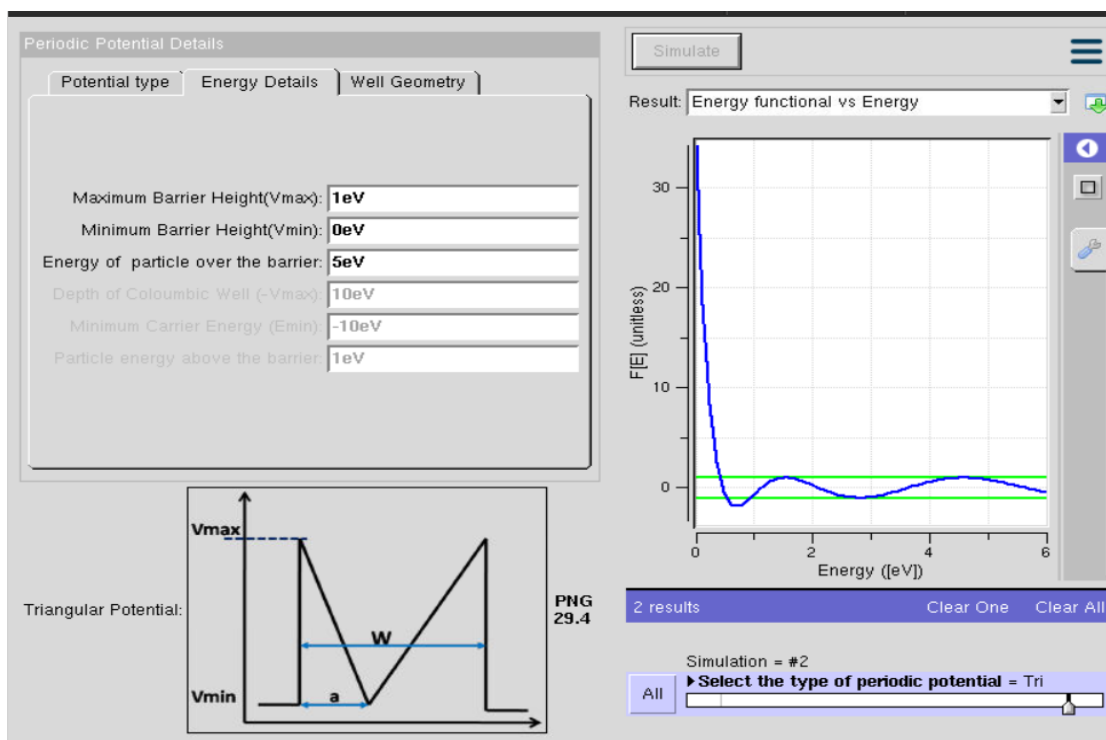
Z Dimension: 5nm



Exp. 3. Simulation of Energy States of Quantum Dot using NanoHub



Case Study 1: Simulate the energy states of cylindrical quantum dots of $X=15$ nm, $Y = 10.5$ nm and $Z = 8$ nm.



Case Study 2: Simulate the energy states of two layer cuboid quantum dots

